Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Financial Time Series I

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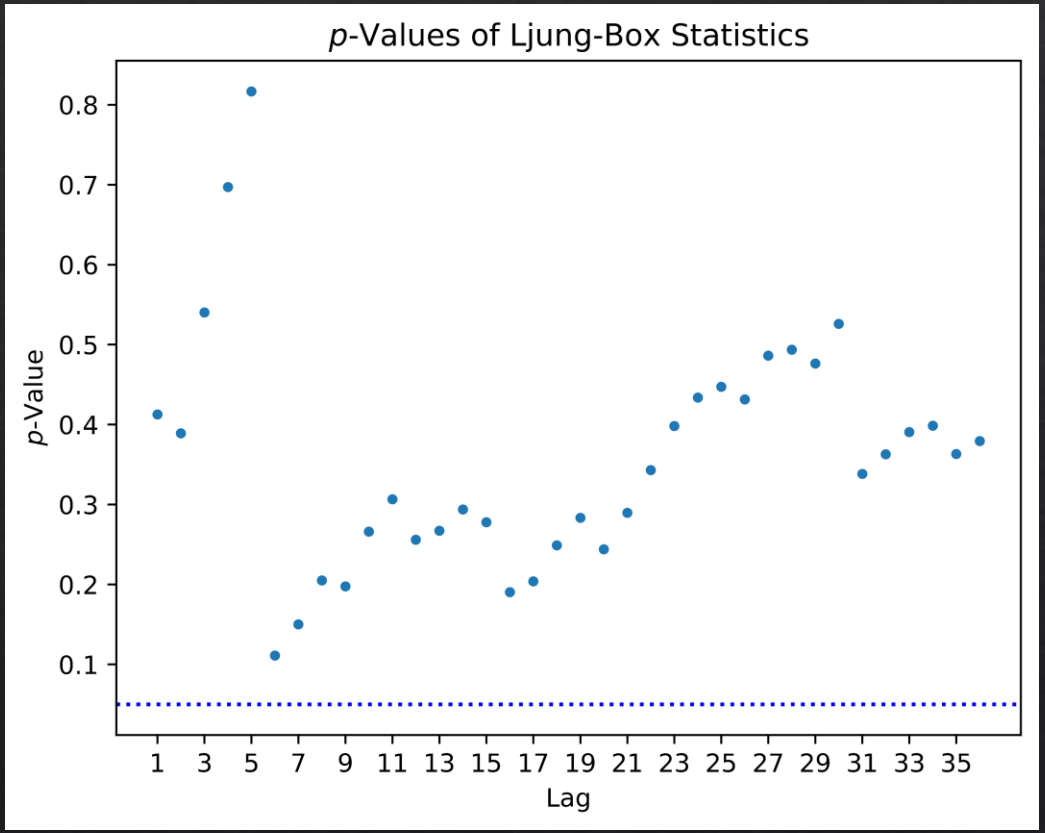
Financial Time Series

- Financial time series analysis has been one of the hottest research topics in recent decades.
- Analyze the stylized facts of financial time series using actual financial data.
- New models different from Box-Jenkins models are needed to characterize these facts.
- For this reason, study the ARCH (Autoregressive Conditional Heteroscedasticity) models and the ARCH effect that changes over time.

ARCH Effect

Monthly Returns of Procter and Gamble Stock from 1961 to 2016

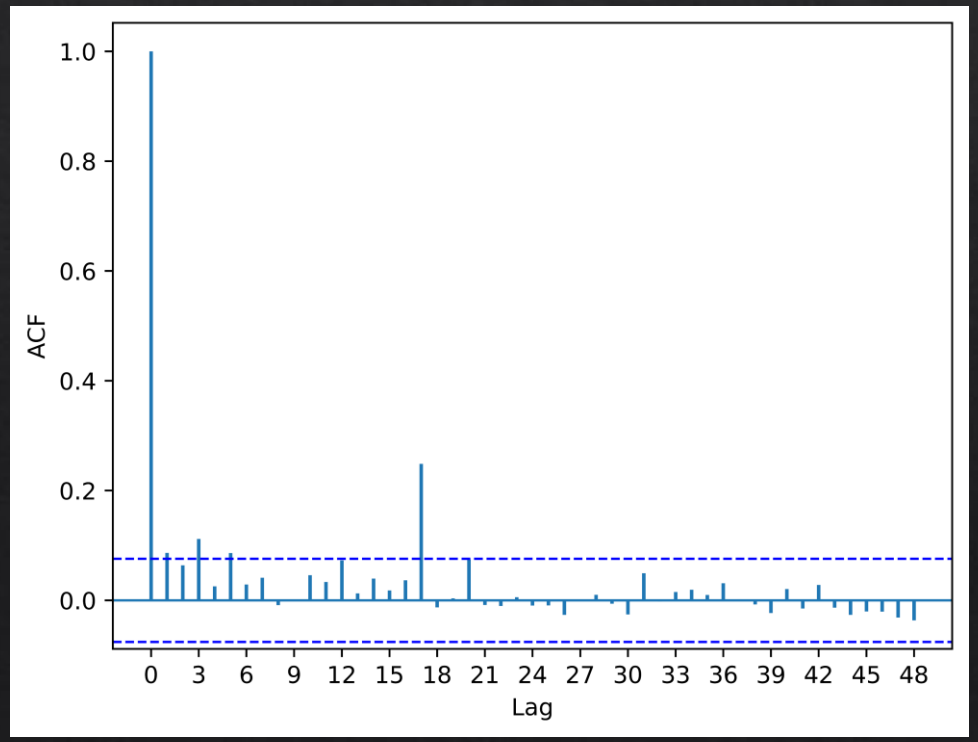
- The return series is a white noise series.
- The *p*-value for the KPSS test of the return series is greater than 0.1, which means that the series is stationary.

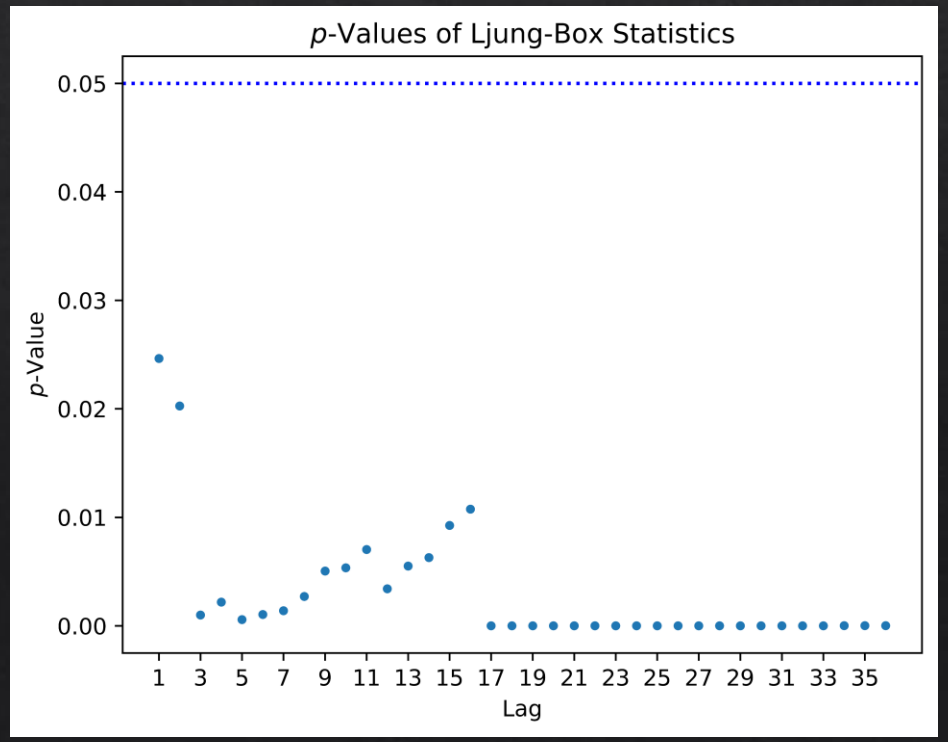


ARCH Effect

Monthly Returns of Procter and Gamble Stock from 1961 to 2016

- The squared return series clearly shows it is autocorrelated.
- The return series has an ARCH effect.

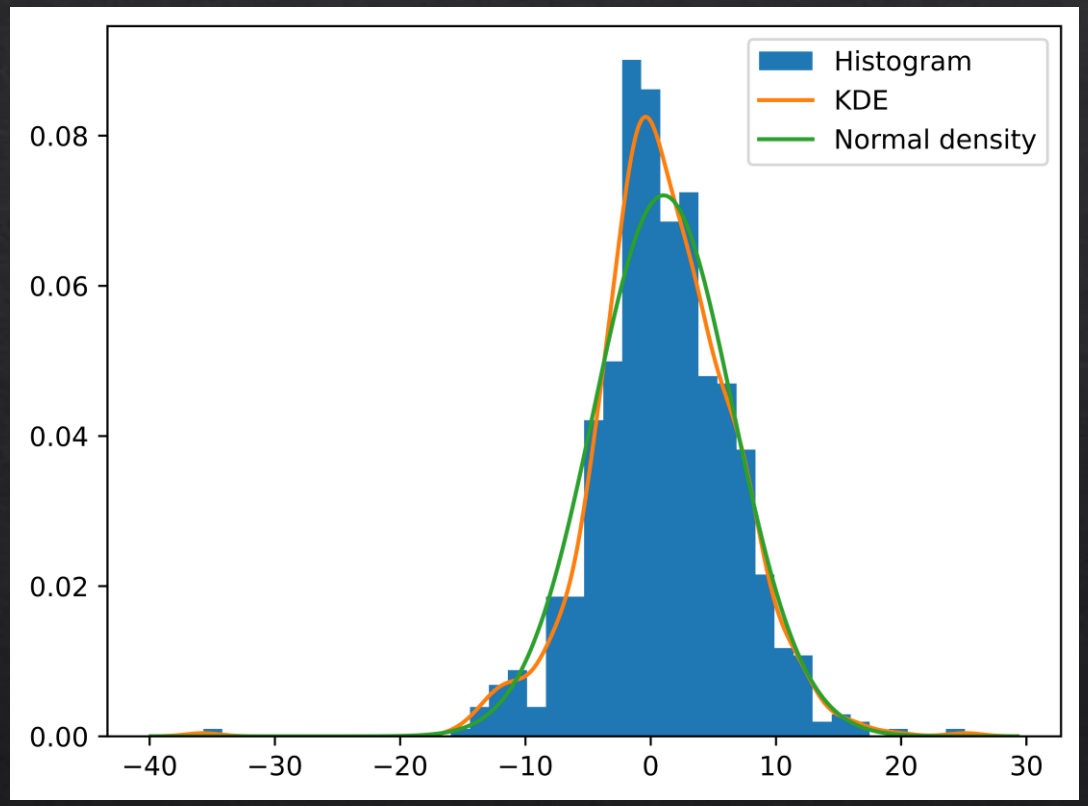




ARCH Effect

Monthly Returns of Procter and Gamble Stock from 1961 to 2016

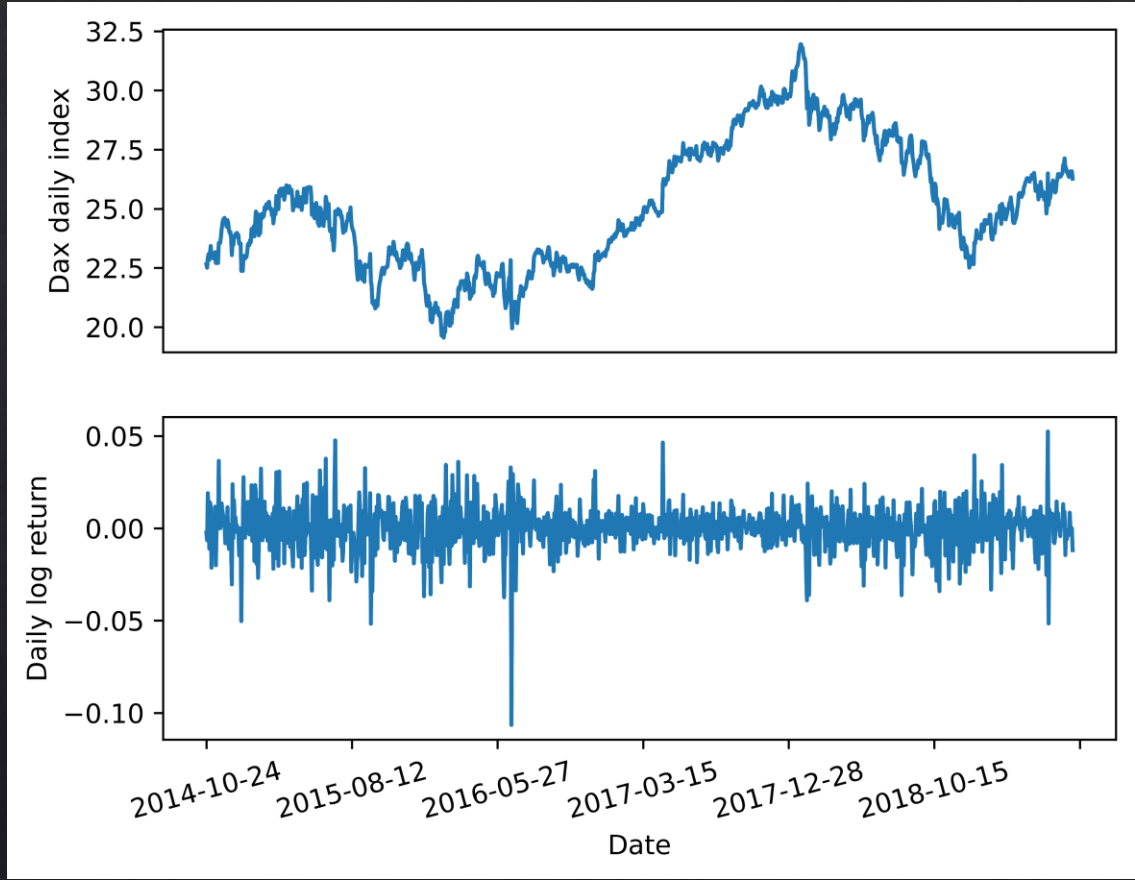
- The distribution density of the return series has a fat (heavy) tail on the left compared with a normal density.
- It is also asymmetric (the symmetric line is the mean=1.0342)



ARCH Effect

Germany DAX Daily Index from Oct 23, 2014, to July 7, 2019

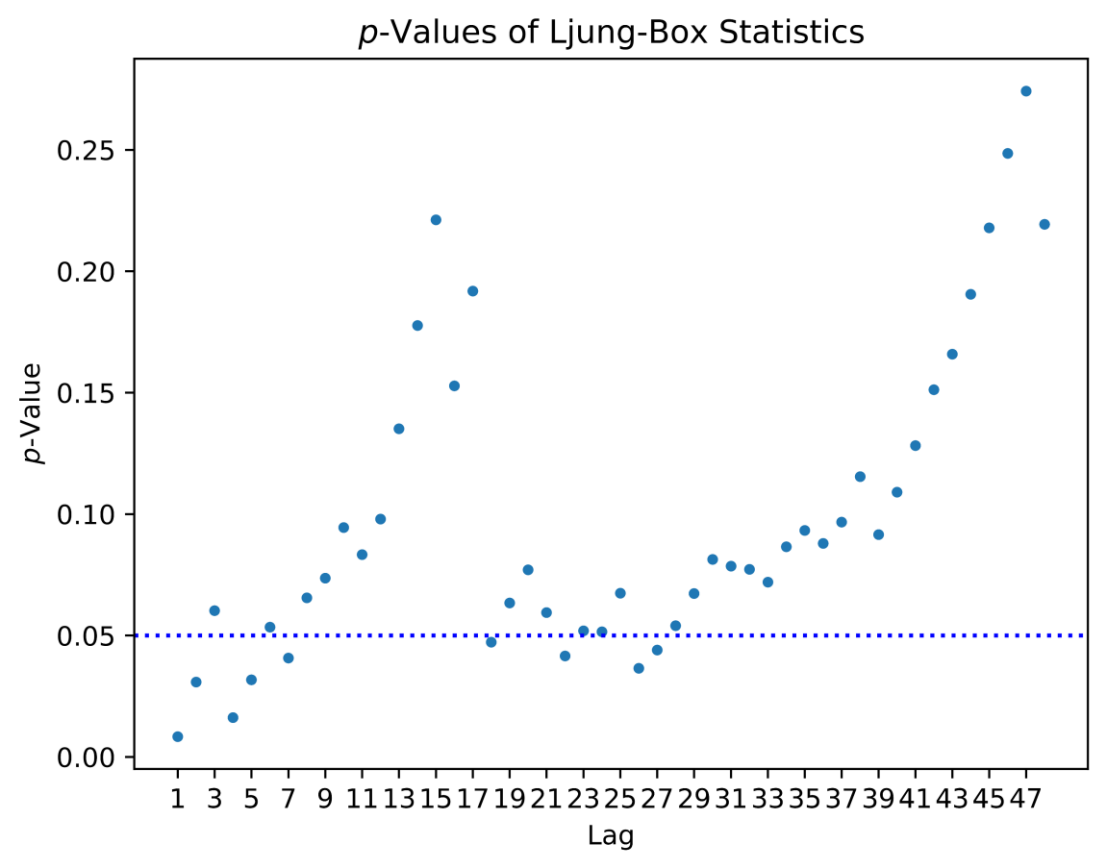
- It is the daily adjusted closing prices from Yahoo!
- Finance data is in the top panel of the figure.
- To study the return series, transform it to the log returns (see bottom panel).



ARCH Effect

Germany DAX Daily Index from Oct 23, 2014, to July 7, 2019

- It is easy to prove that the return series is stationary and has an ARCH effect and fat tails.
- It is not a white noise series.



Stylized Facts of Financial Time Series

Fat (heavy) tails: The distribution density function of returns often has fatter (heavier) tails than the tails of the corresponding normal distribution.

ARCH effect: Although the return series can often be seen as white noise, its squared series may usually be autocorrelated, and these autocorrelations are hardly negative.

Volatility clustering: Large changes in returns tend to cluster in time, and small changes tend to be followed by small changes.

Asymmetry: The distribution of asset returns is slightly negatively skewed.

ARCH Model

- An ARCH(p) model with order p ≥ 1 is of the form

$$\begin{cases} X_t = \sigma_t \varepsilon_t \\ \sigma_t^2 = \omega + \alpha_1 X_{t-1}^2 + \alpha_2 X_{t-2}^2 + \cdots + \alpha_p X_{t-p}^2 \end{cases}$$

  where $\omega \geq 0$, $\alpha_i \geq 0$, and $\alpha_p > 0$ are constants, $\varepsilon_t \sim iid(0,1)$.

- A stochastic process X_t is called an ARCH(p) process if tis satisfies the above equation.

- An ARCH(1) model

$$\begin{cases} X_t = \sigma_t \varepsilon_t \\ \sigma_t^2 = \omega + \alpha_1 X_{t-1}^2 \end{cases}$$

GARCH Model

- A GARCH(p, q) (generalized autoregressive conditional heteroscedasticity) model with order ($p \geq 1$, $q \geq 0$) is of the form

$$\begin{cases} X_t = \sigma_t \varepsilon_t \\ \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i X_{t-1}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2 \end{cases}$$

where $\omega \geq 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, $\alpha_p > 0$ and $\beta_q > 0$ are constants, $\varepsilon_t \sim iid(0,1)$.

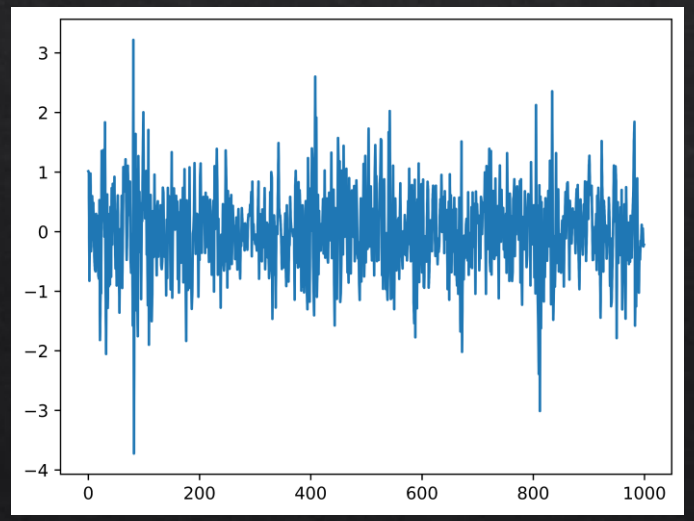
- A stochastic process X_t is called a GARCH(p, q) process if tis satisfies the above equation.

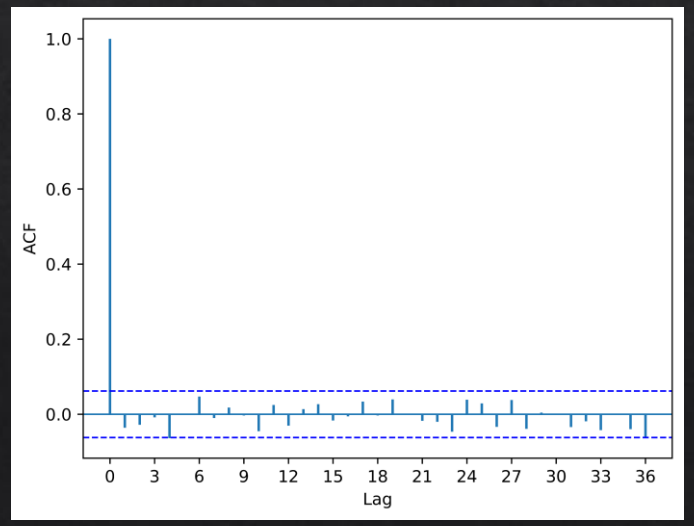
GARCH Model

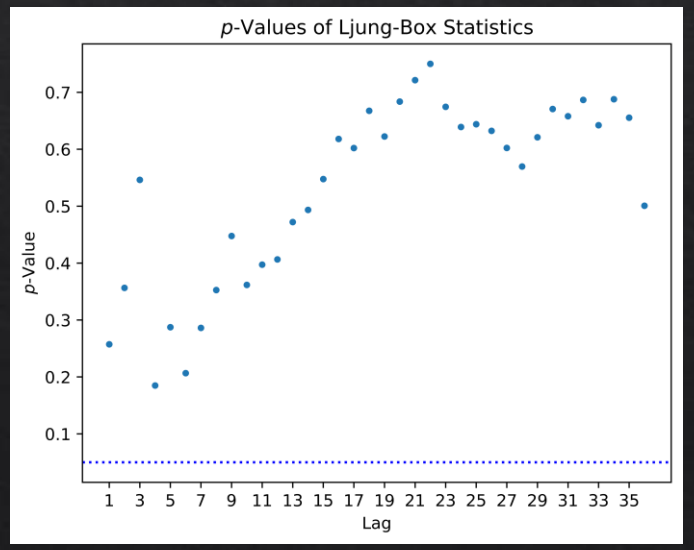
- A GARCH(1,1) model

$$\begin{cases} X_t = \sigma_t \varepsilon_t, \varepsilon_t \sim iid(0,1). \\ \sigma_t^2 = \omega + 0.2X_{t-1}^2 + 0.6\sigma_{t-j}^2 \end{cases}$$

- Left side figure shows the simulated time series of the volatility clustering.
- The ACF and *p*-values shown in figures clearly suggest that the series is a white noise.

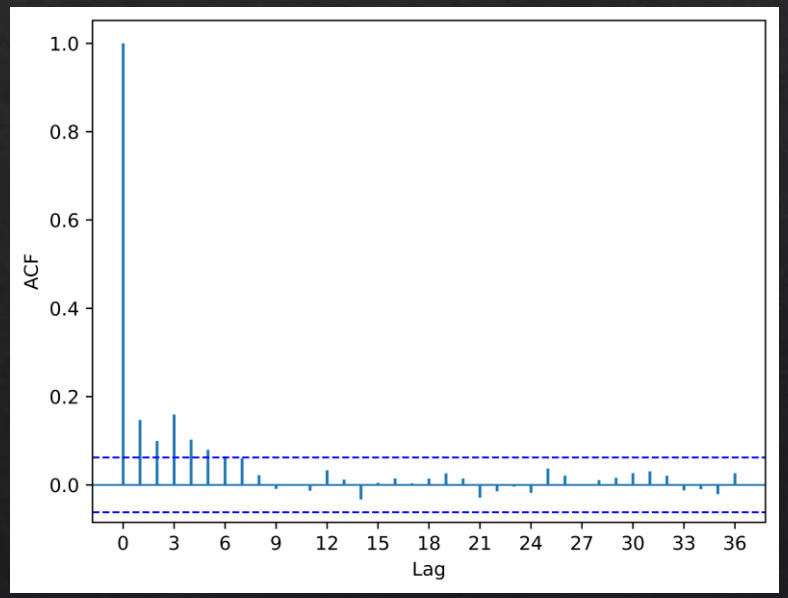


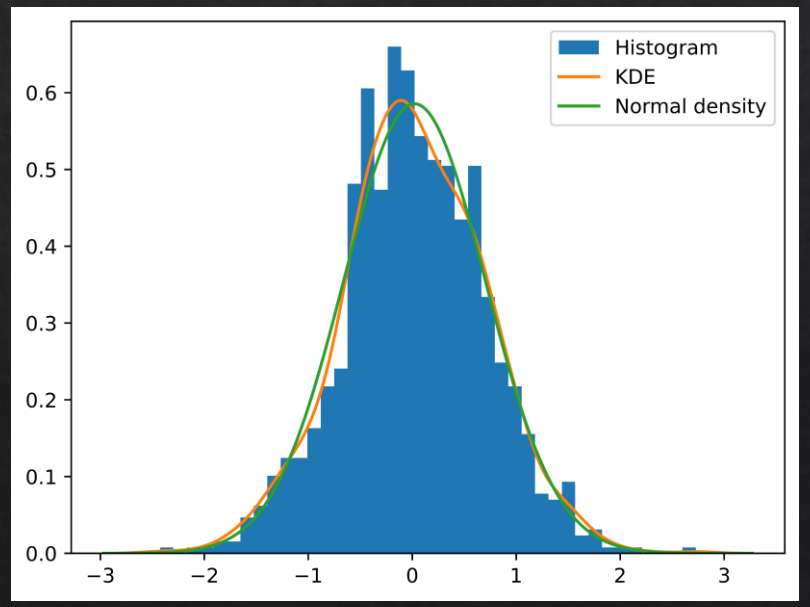




GARCH Model

- ACF plot of the squared simulated data demonstrates that the squared series is autocorrelated or the series has an ARCH effect.
- The left tail of the distribution of the simulated series is heavier than the normal distribution.

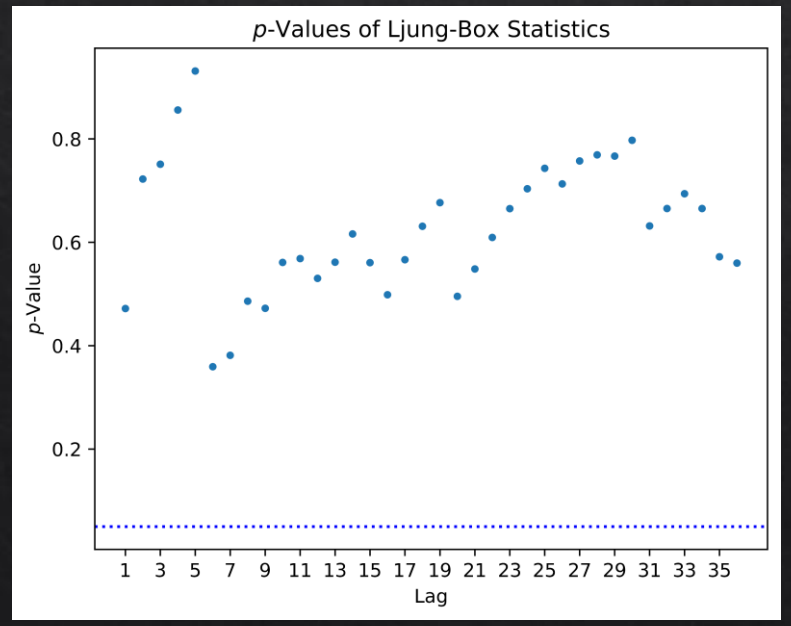


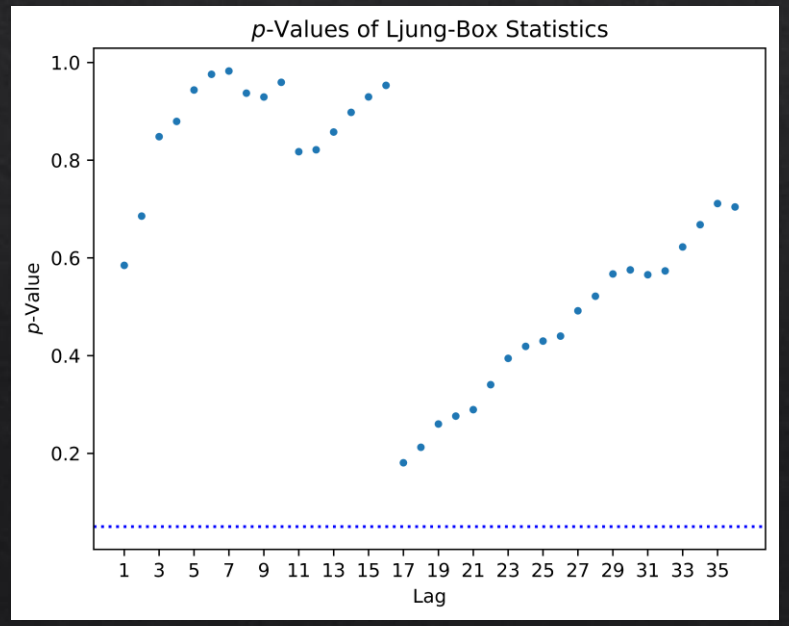


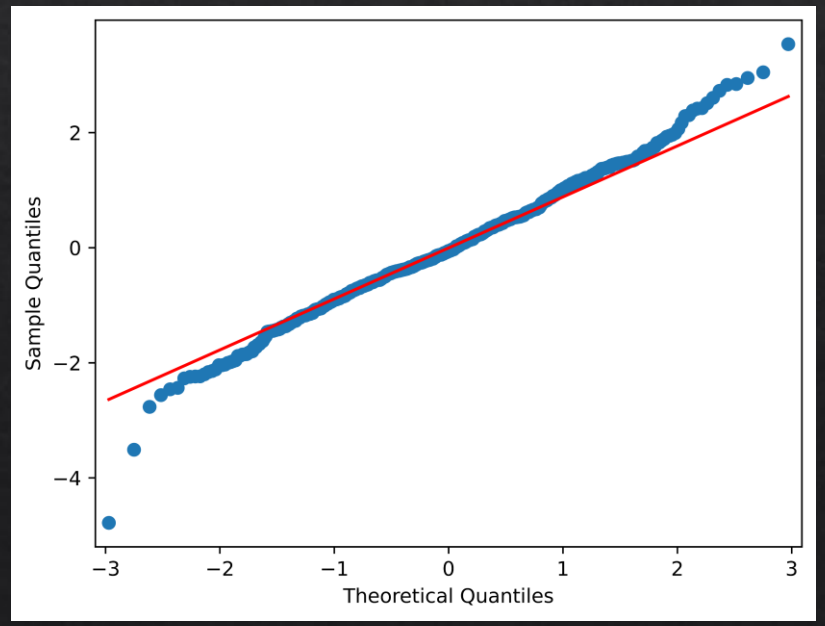
GARCH Model

- A GARCH model continues from PG return data

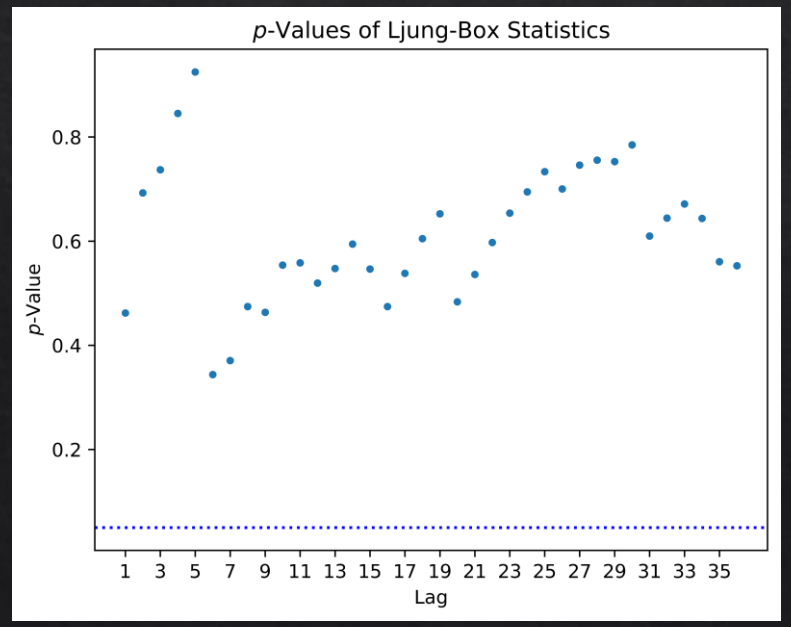
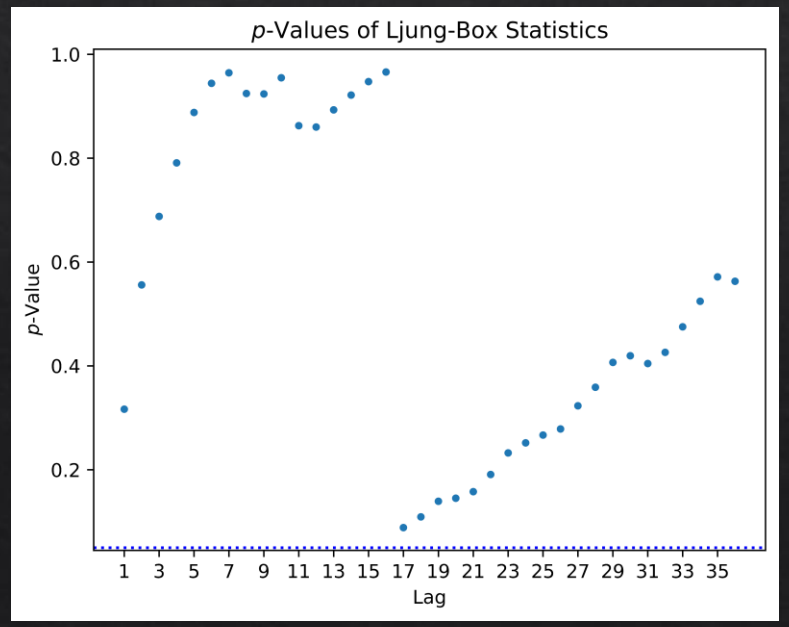
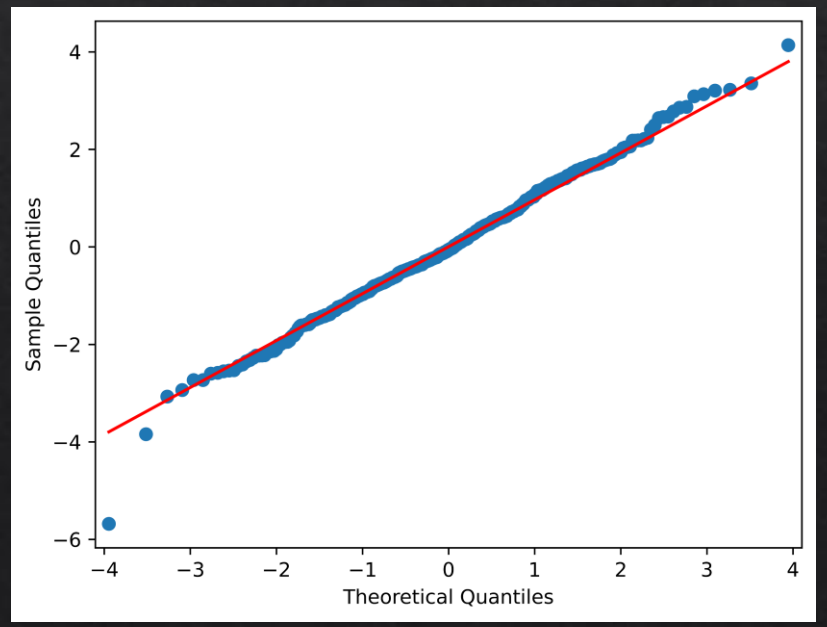
$$\begin{cases} R_t = 1.0918 + X_t \\ X_t = \sigma_t \varepsilon_t, \varepsilon_t \sim iidN(0,1). \\ \sigma_t^2 = 2.8447 + 0.1252 X_{t-1}^2 + 0.7843 \sigma_{t-1}^2 \end{cases}$$







GARCH Model

- A GARCH model

$$\begin{cases} R_t = 1.0157 + X_t \\ X_t = \sigma_t \varepsilon_t, \varepsilon_t \sim iidT(0,1). \\ \sigma_t^2 = 2.7251 + 0.0952 X_{t-1}^2 + 0.8139 \sigma_{t-1}^2 \end{cases}$$